物聯網閘道器—
RFID應用之系統執行

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簡介及系統架構

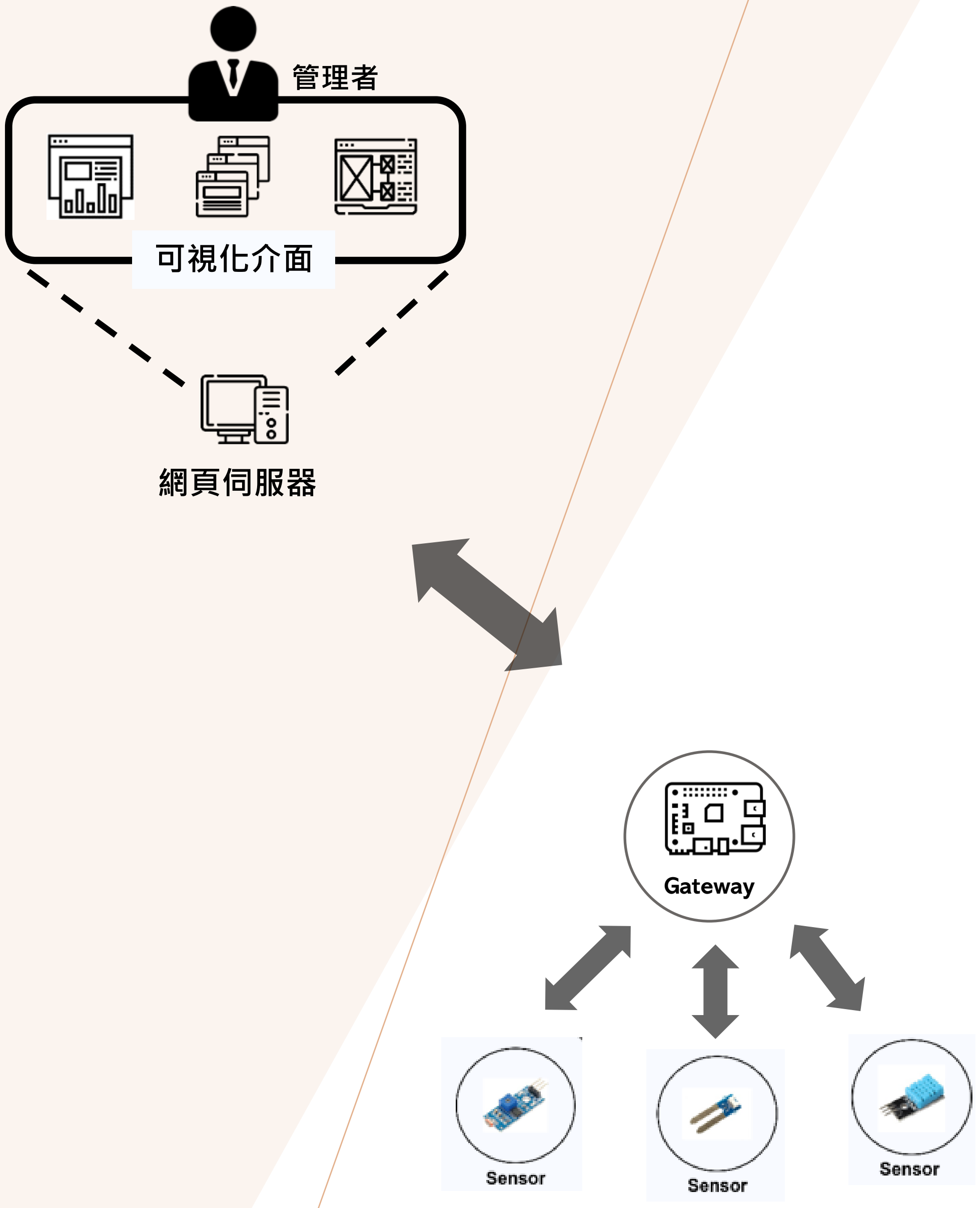
RFID執行流程

硬體目前狀況

閘道器未來優化方向

OUTLINE
大綱

RFID應用之系統架構



物聯網閘道之作用(gateway)

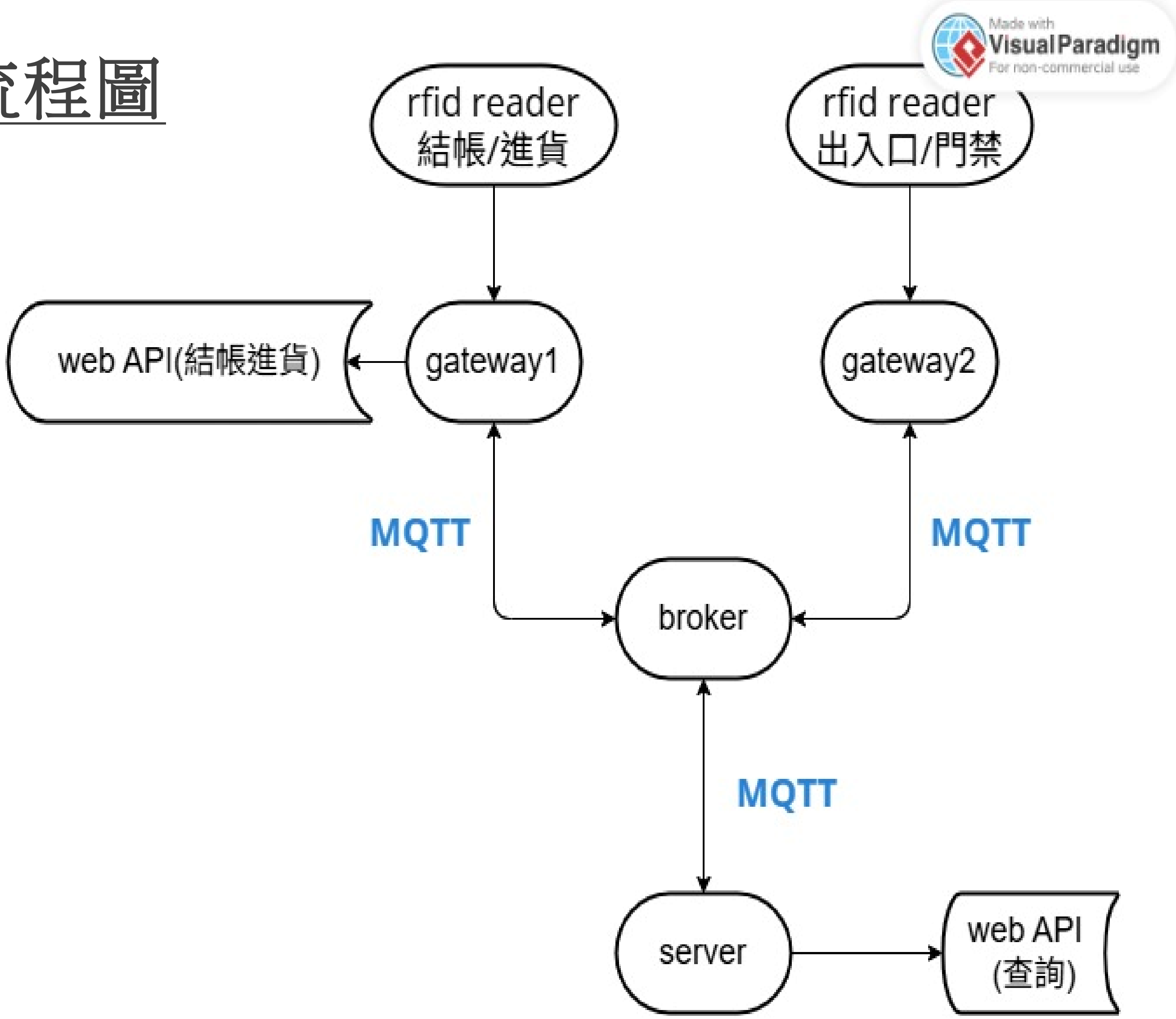
- 物聯網裝置執行任務

  RFID應用中RFID Reader為感測裝置，用來讀取標籤

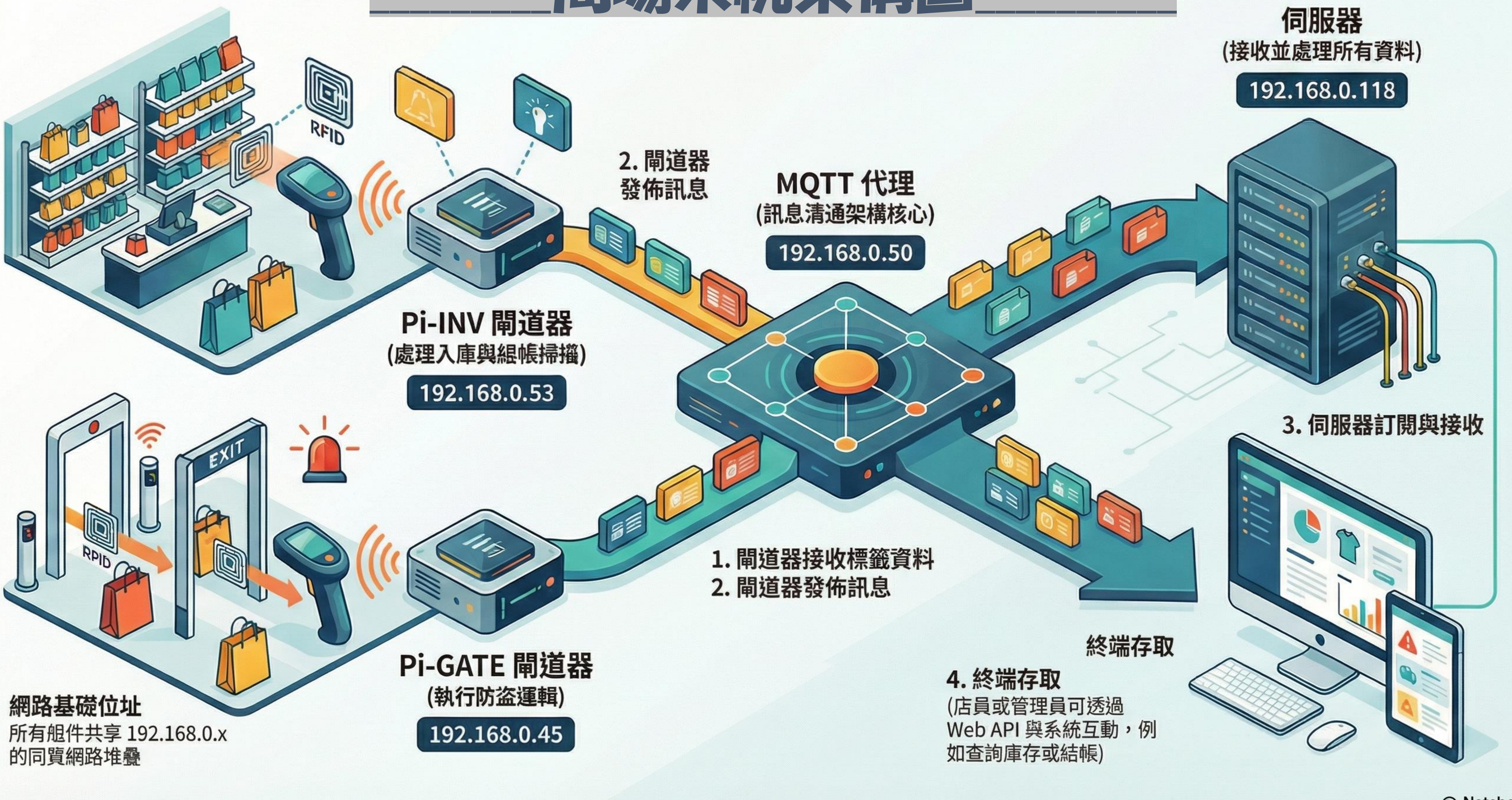
- 閘道器負責把RFID Reader讀到的標籤訊息，轉換並寫入資料庫，再將狀態顯示在系統端，用於結帳與進出口管理。

- 物聯網中可能會有多個閘道器並管理對應裝置

流程圖


rfid reader
結帳/進貨
rfid reader
出入口/門禁
web API(結帳進貨)
gateway1
gateway2
MQTT
MQTT
broker
MQTT
server
web API
(查詢)

商場系統架構圖



伺服器
(接收並處理所有資料)
192.168.0.118
RFID
2. 閘道器
發佈訊息
MQTT 代理
(訊息清通架構核心)
192.168.0.50
Pi-INV 閘道器
(處理入庫與組帳掃描)
192.168.0.53
3. 伺服器訂閱與接收
EXIT
RPID
1. 閘道器接收標籤資料
2. 閘道器發佈訊息
終端存取
Pi-GATE 閘道器
(執行防盜運輯)
192.168.0.45
4. 終端存取
(店員或管理員可透過
Web API 與系統互動，例
如查詢庫存或結帳)
網路基礎位址
所有組件共享 192.168.0.x
的同質網路堆疊

系統執行

一、套件安裝

1. 主機需安裝以下套件：
   - Docker
   - Docker Compose

MacOS (Server)
下載 **Docker Desktop for Mac**並啟動

輸入bash:
docker --version
docker compose version → 確認版本

Linux(gateway-raspberry pi)
安裝**Docker Engine**並啟動
安裝**Docker Compose**（**Compose v2**）並啟動

輸入bash:
docker --version
docker compose version → 確認版本

確保使用者有執行 Docker 的權限!

一、套件安裝

1. 主機需安裝以下套件：

- ➢ Docker
- ➢ Docker Compose

Window

下載 **Docker Desktop for Window**並啟動

輸入cmd:

docker --version

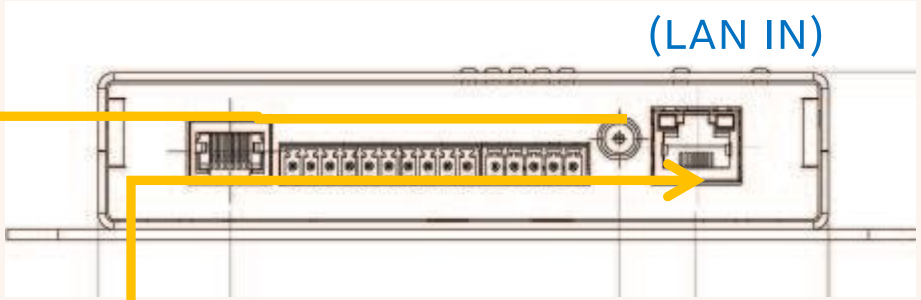
docker compose version → 確認版本

確保使用者有執行 **Docker** 的權限!

```
[john@raspberrypi:~/GW2 $ docker --version
Docker version 29.1.3, build f52814d
```

二、硬體設置(RFID reader)

1. AR70 UHF RFID Reader(出入口)接線方式



電源線
規格:24V以上/1.5A

接網路線

接到reader天線上

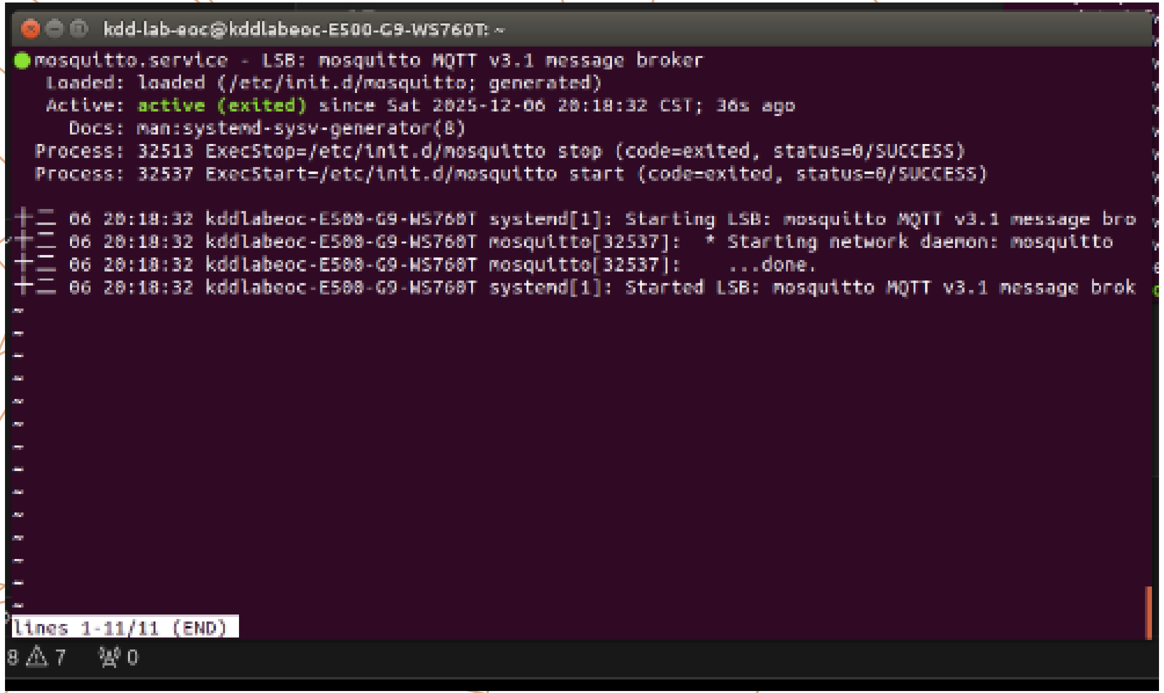
二、硬體設置(RFID reader)

1. AR70 UHF RFID Reader基礎設定

- 動態IP:192.168.1.85
- 帳號keonn
- 密碼b94fthWDdN7uELc/eib3emsaJJANdCNOlVvw9Y/WA4E(原廠重設)
- Port:

1. 把reader調成Ethernet / TCP Server模式
2. Bash輸入:sudo netstat -tulpn | grep LISTEN，找到接收中的port
3. 執行read_rfid.py

三、MQTT環境設定-Broker



將server_config.ini的IP改成Broker(例如本專案是192.168.1.50)

1. 安裝mosquitto

Bash

sudo apt install mosquitto mosquitto-clients –y

2. 設定防火牆

允許外部網路通過 1883 Port

Bash:sudo ufw allow 1883

3. 修改mosquitto設定檔、啟動mosquitto

- 新增一個設定檔來允許外部連線→

Bash:sudo nano /etc/mosquito/conf.d/external.conf

- 複製貼上以下設定:

{ listener 1883
allow_anonymous true }

- **重啟後確認mosquitto有亮綠燈**

四、環境準備

1. 確認網路環境可連線至 MQTT Broker(啟動Mosquitto後測試)

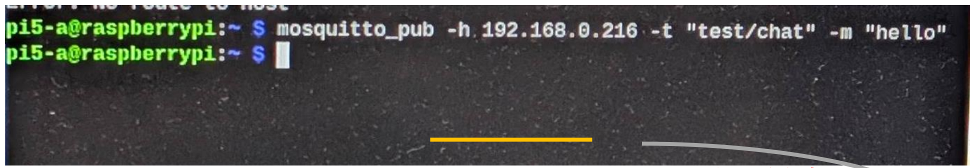
<u>要在同一個子網路內192.168.0.XX(前三碼一樣)</u>

找brokerIP

監聽中

回應

測試發送訊息



gateway(發送端)

依實際broker IP

2. 確認專案目錄結構完整

```
general_gateway/
├──── apps/
│    └──── inventory/
│         ├──── app_executor_inventory.py
│         └──── db_inventory.py
│         └──── models /
│              └──── enum/task_type.py
├── services/
│    ├──── doorkeeper.py
│    └──── task_manager.py
├──── fleet_server.sql
├── fleet_gateway.sql
├──── server_config.ini
└──── gateway_config.ini
```

（app/templates/Index.html）Web API(操作頁面)

五、執行docker

⇨ 終端機

1. 進入資料夾:cd 專案資料夾/
2. 執行docker up
3. 執行docker compose exec -T mysql mysql -uroot -proot < fleet_gateway(server).sql啟動資料庫
4. 進入專案根目錄:cd general_gateway/

4. 啟動 Docker Compose：

Server:`docker compose up -d server mysql redis`

Gateway:`docker compose up -d gateway mysql redis`

此指令會同時啟動以下容器：

- MySQL（資料庫）
- Redis（任務狀態與同步控制）
- Server 端服務/Gateway 端服務

六、資料庫初始化

1.MySQL 容器啟動後，系統會自動執行初始化 SQL：

A. 建立 Product_Item、Operation_Log、 gate_Log 等資料表

B. 若資料表已存在，則刪掉重建

C. 確認資料庫成功啟動：`docker logs iot_mysql`

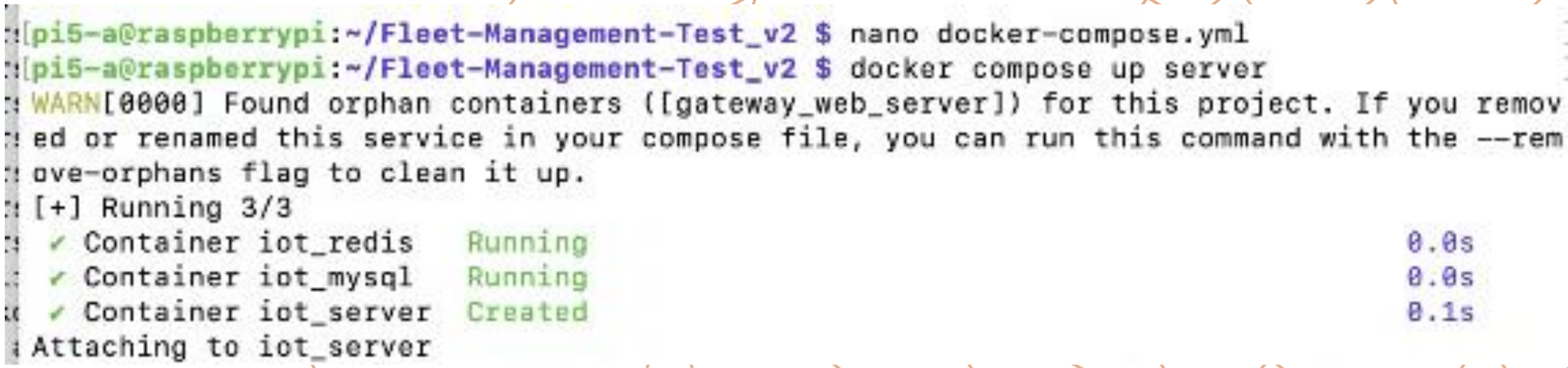
七、確認 GATEWAY 與 SERVER 啟動狀態

1.查看目前執行中的容器：

Bash：`docker ps`

2.確認以下服務皆為啟動狀態：

- iot_mysql
- iot_redis
- iot_gateway/iot_server



3.查看 Gateway log，確認服務已正常啟動並連線 MQTT：

Bash：`docker logs -f iot_gateway/server`

八、確認網頁與 API 服務

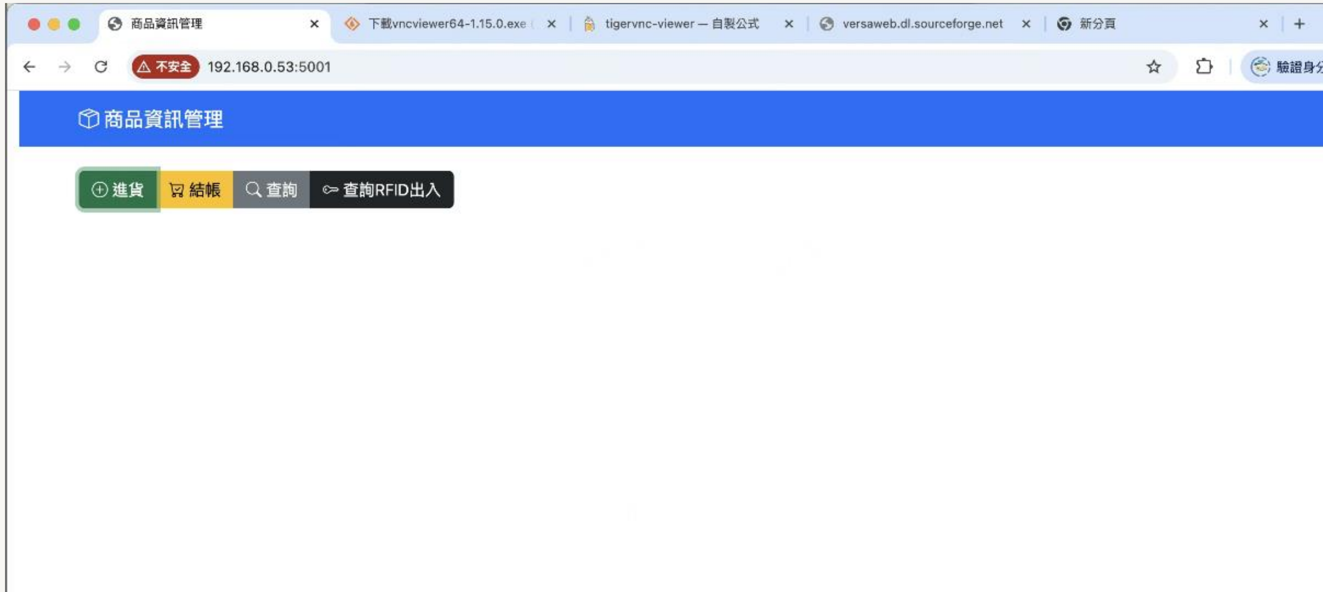
1. Gateway Web 介面：

➢瀏覽器開啟 http://<Gateway_IP>:5001

2. Server Web / API 介面：

➢瀏覽器開啟 http://<Server_IP>:5050

3. 若出現畫面或 API 回應，即代表系統啟動成功



九、常見錯誤

檢查層級	常見情況	可能原因	排除步驟
L1 網路層	No route to host Destination Unreachable	裝置位於不同網段或 Wi-Fi 斷線	1. 檢查 IP 前三碼是否相同 (192.168.1.x)
			2. 使用 ping 指令確認互通性
L2 服務層	Connection Refused Docker/容器一直 Restarting	服務未啟動、Port 被佔用	1. docker ps -a 檢查容器狀態
			2. systemctl status mosquitto
			3. 檢查 docker-compose.yml Port 設定
	資料庫連線失敗 (MySQL Error)	帳號密碼錯誤、還沒初始化	1. docker logs iot_mysql 查看log
			2. 確認 gateway_config.ini 帳密正確
L3 資料層	MQTT 有連線但收不到訊息	Topic 設定不一致	1. 檢查 Reader 發送的 Topic 字串
			2. 檢查 Gateway 訂閱的 Topic 字串是否完全相同
	Gateway 收到資料但 Server 沒顯示	JSON 格式錯誤、API 失敗	1. 在 Gateway 印出收到的 Payload (Print Debug)
			2. 確認 JSON 欄位名稱 (如 tag_uid) 與後端一致

硬體現況

1. Docker

兩台樹莓派上跟server的docker都測試過，可正常啟動，資料表可自動建立

2. MQTT

Broker已架好，啟動後即可由樹梅派GW02發送訊息到broker，broker也能再發至server上，Server回覆收到

Server端docker啟動成功畫面

```
(base) kddlab@kdddeMBP general_gateway % docker compose up -d server mysql redis
[+] Running 3/3
 ✔ Container iot_redis    Running                                              0.0s
 ✔ Container iot_mysql    Running                                              0.0s
 ✔ Container iot_server   Started                                              0.1s
(base) kddlab@kdddeMBP general_gateway % docker logs iot_mysql
2025-12-10 07:26:51+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.0.44-1.el9 started.
2025-12-10 07:26:51+00:00 [Note] [Entrypoint]: Switching to dedicated user 'mysql'
2025-12-10 07:26:51+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.0.44-1.el9 started.
'/var/lib/mysql/mysql.sock' -> '/var/run/mysqld/mysqld.sock'
2025-12-10T07:26:52.187219Z 0 [Warning] [MY-011068] [Server] The syntax '--skip-host-cache' is deprecated and will be removed in a future release. Please use SET GLOBAL host_cache_size=0 instead.
2025-12-10T07:26:52.195500Z 0 [System] [MY-010116] [Server] /usr/sbin/mysqld (mysqld 8.0.44) starting as process 1
2025-12-10T07:26:52.197111Z 0 [Warning] [MY-010159] [Server] Setting lower_case_table_names=2 because file system for /var/lib/mysql/ is case insensitive
2025-12-10T07:26:52.202129Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
2025-12-10T07:26:52.583603Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
2025-12-10T07:26:52.797488Z 4 [System] [MY-013381] [Server] Server upgrade from '80042' to '80044' started.
2025-12-10T07:26:54.612094Z 4 [System] [MY-013381] [Server] Server upgrade from '80042' to '80044' completed.
2025-12-10T07:26:54.630067Z 0 [System] [MY-010229] [Server] Starting XA crash recovery...
2025-12-10T07:26:54.641161Z 0 [System] [MY-010232] [Server] XA crash recovery finished.
2025-12-10T07:26:54.657091Z 0 [Warning] [MY-010068] [Server] CA certificate ca.pem is self signed.
2025-12-10T07:26:54.657113Z 0 [System] [MY-013602] [Server] Channel mysql_main configured to support TLS. Encrypted connections are now supported for this channel.
2025-12-10T07:26:54.658759Z 0 [Warning] [MY-011810] [Server] Insecure configuration for --pid-file: Location '/var/run/mysqld' in the path is accessible to all OS users. Consider choosing a different dire
ctory.
2025-12-10T07:26:54.664622Z 0 [System] [MY-011323] [Server] X Plugin ready for connections. Bind-address: '::' port: 33060, socket: /var/run/mysqld/mysqlx.sock
2025-12-10T07:26:54.664662Z 0 [System] [MY-010931] [Server] /usr/sbin/mysqld: ready for connections. Version: '8.0.44'  socket: '/var/run/mysqld/mysqld.sock'  port: 3306  MySQL Community Server - GPL.
2026-01-08 08:42:59+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.0.44-1.el9 started.
2026-01-08 08:43:00+00:00 [Note] [Entrypoint]: Switching to dedicated user 'mysql'
2026-01-08 08:43:00+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.0.44-1.el9 started.
'/var/lib/mysql/mysql.sock' -> '/var/run/mysqld/mysqld.sock'
2026-01-08T08:43:01.048651Z 0 [Warning] [MY-011068] [Server] The syntax '--skip-host-cache' is deprecated and will be removed in a future release. Please use SET GLOBAL host_cache_size=0 instead.
2026-01-08T08:43:01.054857Z 0 [System] [MY-010116] [Server] /usr/sbin/mysqld (mysqld 8.0.44) starting as process 1
2026-01-08T08:43:01.056174Z 0 [Warning] [MY-010159] [Server] Setting lower_case_table_names=2 because file system for /var/lib/mysql/ is case insensitive
2026-01-08T08:43:01.062487Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
2026-01-08T08:43:01.450389Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
2026-01-08T08:43:01.575733Z 0 [System] [MY-010229] [Server] Starting XA crash recovery...
2026-01-08T08:43:01.607886Z 0 [System] [MY-010232] [Server] XA crash recovery finished.
2026-01-08T08:43:01.688875Z 0 [Warning] [MY-010068] [Server] CA certificate ca.pem is self signed.
2026-01-08T08:43:01.688906Z 0 [System] [MY-013602] [Server] Channel mysql_main configured to support TLS. Encrypted connections are now supported for this channel.
2026-01-08T08:43:01.690690Z 0 [Warning] [MY-011810] [Server] Insecure configuration for --pid-file: Location '/var/run/mysqld' in the path is accessible to all OS users. Consider choosing a different dire
ctory.
2026-01-08T08:43:01.736796Z 0 [System] [MY-011323] [Server] X Plugin ready for connections. Bind-address: '::' port: 33060, socket: /var/run/mysqld/mysqlx.sock
2026-01-08T08:43:01.736851Z 0 [System] [MY-010931] [Server] /usr/sbin/mysqld: ready for connections. Version: '8.0.44'  socket: '/var/run/mysqld/mysqld.sock'  port: 3306  MySQL Community Server - GPL.
2026-01-09 12:25:54+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.0.44-1.el9 started.
2026-01-09 12:25:54+00:00 [Note] [Entrypoint]: Switching to dedicated user 'mysql'
2026-01-09 12:25:54+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.0.44-1.el9 started.
'/var/lib/mysql/mysql.sock' -> '/var/run/mysqld/mysqld.sock'
2026-01-09T12:25:54.572263Z 0 [Warning] [MY-011068] [Server] The syntax '--skip-host-cache' is deprecated and will be removed in a future release. Please use SET GLOBAL host_cache_size=0 instead.
2026-01-09T12:25:54.574048Z 0 [System] [MY-010116] [Server] /usr/sbin/mysqld (mysqld 8.0.44) starting as process 1
2026-01-09T12:25:54.575356Z 0 [Warning] [MY-010159] [Server] Setting lower_case_table_names=2 because file system for /var/lib/mysql/ is case insensitive
2026-01-09T12:25:54.582731Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
2026-01-09T12:25:54.855078Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
2026-01-09T12:25:54.959960Z 0 [System] [MY-010229] [Server] Starting XA crash recovery...
2026-01-09T12:25:54.965524Z 0 [System] [MY-010232] [Server] XA crash recovery finished.
2026-01-09T12:25:54.992610Z 0 [Warning] [MY-010068] [Server] CA certificate ca.pem is self signed.
2026-01-09T12:25:54.992644Z 0 [System] [MY-013602] [Server] Channel mysql_main configured to support TLS. Encrypted connections are now supported for this channel.
2026-01-09T12:25:54.994322Z 0 [Warning] [MY-011810] [Server] Insecure configuration for --pid-file: Location '/var/run/mysqld' in the path is accessible to all OS users. Consider choosing a different dire
ctory.
2026-01-09T12:25:55.010164Z 0 [System] [MY-011323] [Server] X Plugin ready for connections. Bind-address: '::' port: 33060, socket: /var/run/mysqld/mysqlx.sock
2026-01-09T12:25:55.010206Z 0 [System] [MY-010931] [Server] /usr/sbin/mysqld: ready for connections. Version: '8.0.44'  socket: '/var/run/mysqld/mysqld.sock'  port: 3306  MySQL Community Server - GPL.
```

DOCKER運行畫面

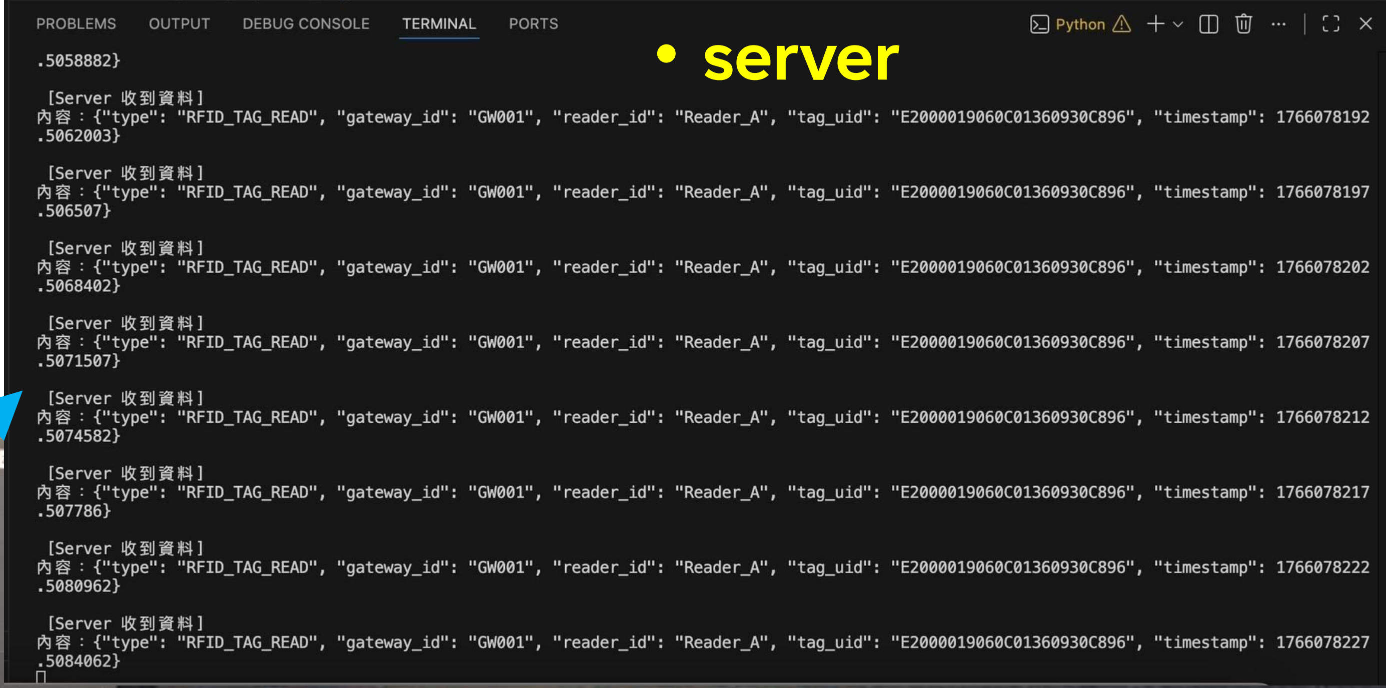
樹莓派GW02

```
pi5-a@raspberrypi:~/Fleet-Management-Test_v2 $ nano docker-compose.gateway.yml
pi5-a@raspberrypi:~/Fleet-Management-Test_v2 $ nano docker-compose.yml
pi5-a@raspberrypi:~/Fleet-Management-Test_v2 $ docker compose up gateway
WARN[0000] Found orphan containers ([gateway_web_server]) for this project. If you remov
ed or renamed this service in your compose file, you can run this command with the --rem
ove-orphans flag to clean it up.
[+] Running 4/4
 ✔ Network fleet-management-test_v2_default  Created                          0.0s
 ✔ Container iot_redis                       Crea...                          0.2s
 ✔ Container iot_mysql                       Crea...                          0.2s
 ✔ Container iot_gateway                     Cr...                            0.1s
Attaching to iot_gateway
iot_gateway  | ---Config---
iot_gateway  | ROLE: Gateway
iot_gateway  | SERVER_ID: 140.116.39.171
iot_gateway  | DELIVER_WAY: Ethernet
iot_gateway  | ---Config---
iot_gateway  | Creating new instance: <general_gateway.main.GeneralGateway object at 0x7
fff00517fd0>
iot_gateway  | general_gateway doorkeeper
iot_gateway  | doorkeeper: start
iot_gateway  | Connect MQTT Broker Successfully!!!
iot_gateway  |  * Serving Flask app 'app' (lazy loading)
iot_gateway  |  * Environment: production
iot_gateway  |    WARNING: This is a development server. Do not use it in a production d
eployment.
iot_gateway  |    Use a production WSGI server instead.
iot_gateway  |  * Debug mode: off
iot_gateway  |  * Running on all addresses (0.0.0.0)
iot_gateway  |    WARNING: This is a development server. Do not use it in a production d
eployment.
iot_gateway  |  * Running on http://127.0.0.1:5000
iot_gateway  |  * Running on http://172.22.0.4:5000 (Press CTRL+C to quit)
iot_gateway  | _deal_task: has_task 0
iot_gateway  | _deal_task: doing_task 0
iot_gateway  | _deal_task: has_task 0
iot_gateway  | _deal_task: doing_task 0
iot_gateway  | _deal_task: has_task 0
iot_gateway  | _deal_task: doing_task 0
iot_gateway  | _deal_task: has_task 0
iot_gateway  | _deal_task: doing_task 0
iot_gateway  | _deal_task: has_task 0
iot_gateway  | _deal_task: doing_task 0
iot_gateway  | _deal_task: has_task 0
```

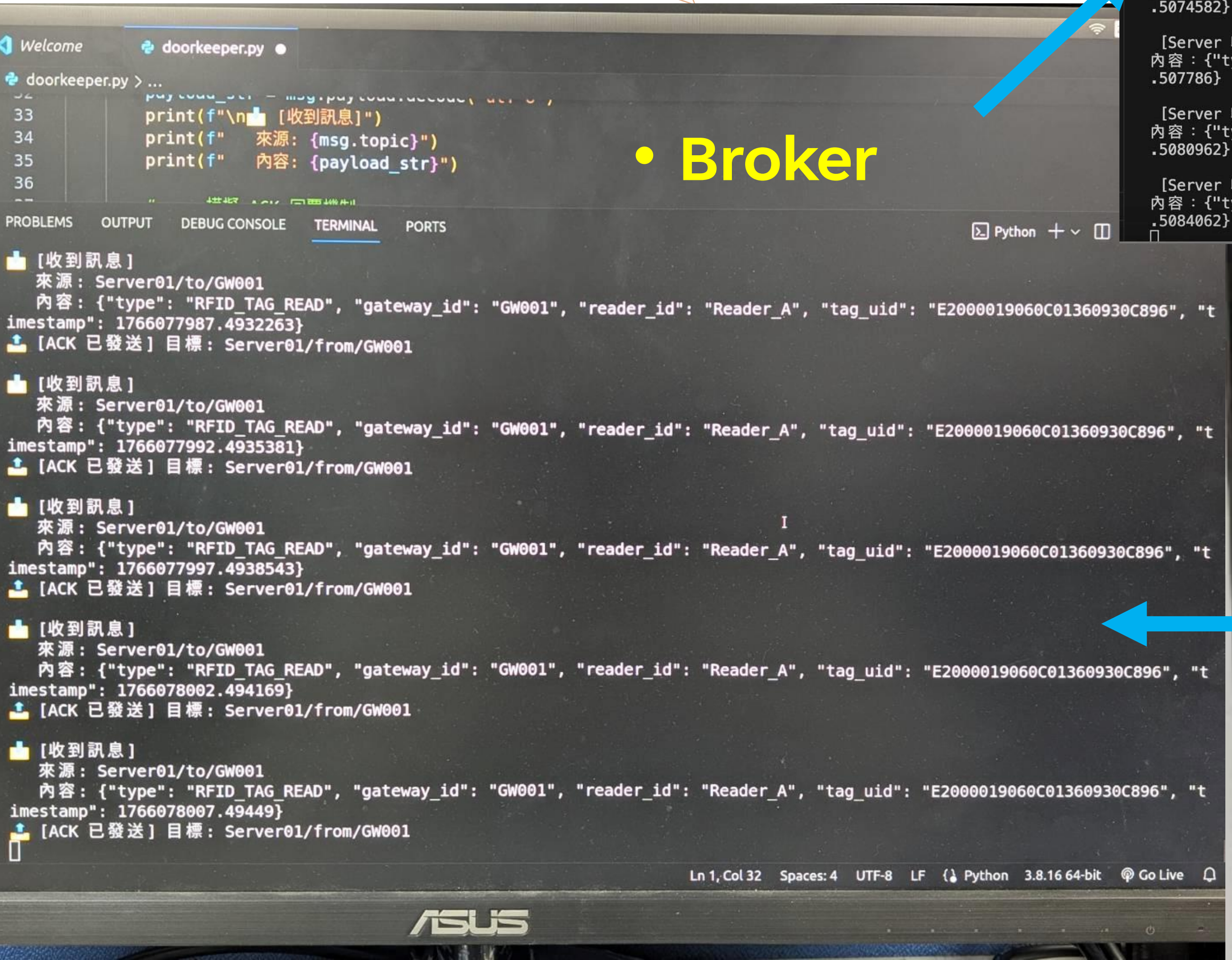
樹莓派GW01

```
john@raspberrypi:~/GW2 $ docker compose up gateway
[+] up 1/1
 ✔ Container iot_redis Running                                               0.0s
Attaching to iot_gateway
iot_gateway  | ---Config---
iot_gateway  | ROLE: Gateway
iot_gateway  | SERVER_ID: 140.116.39.171
iot_gateway  | DELIVER_WAY: Ethernet
iot_gateway  | ---Config---
iot_gateway  | Creating new instance: <general_gateway.main.GeneralGateway object at 0
x7fff11feafd0>
iot_gateway  | general_gateway doorkeeper
iot_gateway  | doorkeeper: start
iot_gateway  | Connect MQTT Broker Successfully!!!
iot_gateway  |  * Serving Flask app 'app' (lazy loading)
iot_gateway  |  * Environment: production
iot_gateway  |    WARNING: This is a development server. Do not use it in a production
 deployment.
iot_gateway  |    Use a production WSGI server instead.
iot_gateway  |  * Debug mode: off
iot_gateway  |  * Running on all addresses (0.0.0.0)
iot_gateway  |    WARNING: This is a development server. Do not use it in a production
 deployment.
iot_gateway  |  * Running on http://127.0.0.1:5000
iot_gateway  |  * Running on http://172.18.0.3:5000 (Press CTRL+C to quit)
john@raspberrypi:~/GW2 $ cd ..
john@raspberrypi:~ $ \hostname -I
192.168.1.42 172.17.0.1 172.18.0.1
john@raspberrypi:~ $ docker compose up gateway
no configuration file provided: not found
john@raspberrypi:~ $ cd GW2
john@raspberrypi:~/GW2 $ docker compose up gateway
[+] up 2/2
 ✔ Container iot_redis    Running                                            0.0s
 ✔ Container iot_gateway  Running                                            0.0s
Attaching to iot_gateway
iot_gateway  | _deal_task: has_task 0
iot_gateway  | _deal_task: doing_task 0
iot_gateway  | _deal_task: has_task 0
iot_gateway  | _deal_task: doing_task 0
iot_gateway  | _deal_task: has_task 0
iot_gateway  | _deal_task: doing_task 0
iot_gateway  | _deal_task: has_task 0
iot_gateway  | _deal_task: doing_task 0
```

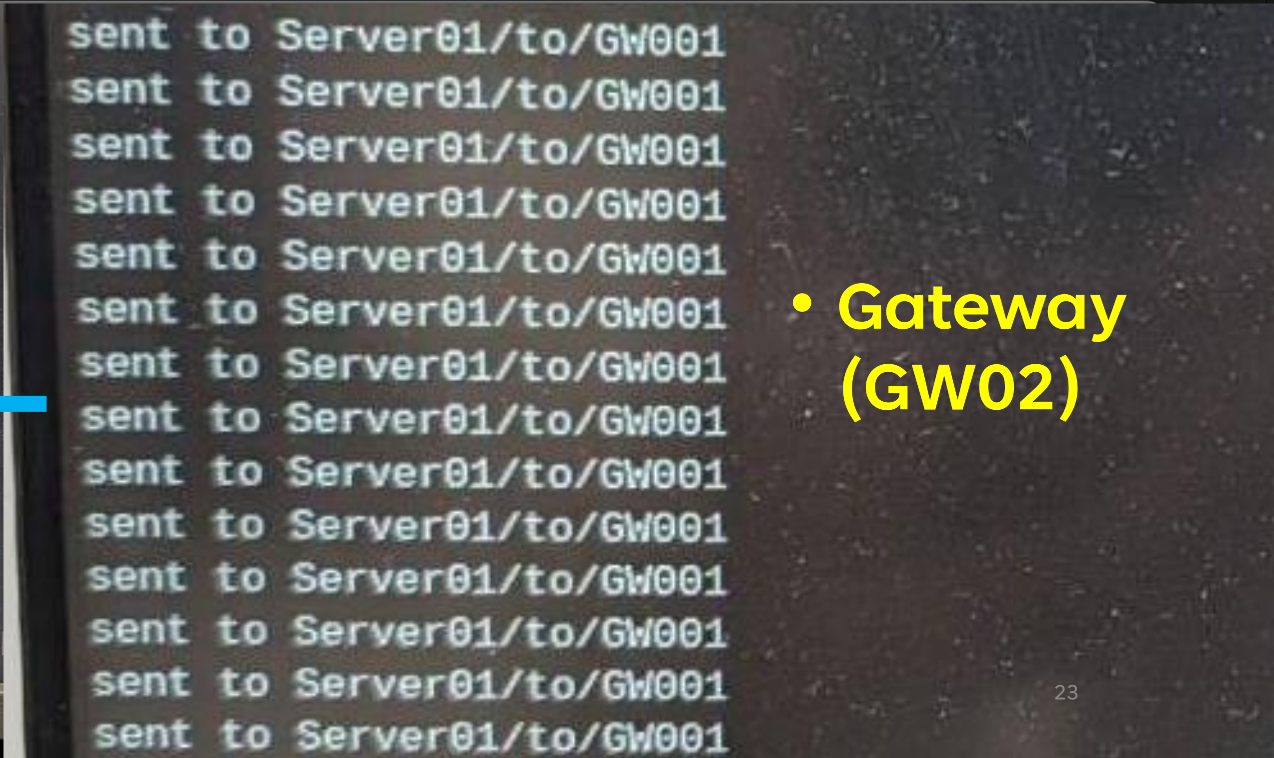
MQTT運行畫面



- server



- Broker



- Gateway (GW02)

RFID reader

AR70 UHF RFID Reader (出入口)

- 已經找到IP:192.168.1.85

(是動態IP，只要關掉就要讓樹梅派重新發IP給他，然後重新把IP掃出來，目前還沒有找到怎麼改成固定IP)

- Console進的去了 (http://192.168.1.85:8080)
- port=3161能通，但還有一些reader模式設定的問題，所以暫時還讀不到tag

SparkFun Simultaneous RFID Reader(結帳)

還沒開始試

閘道器未來優化方向

MQTT優化

帳號權限驗證

每一個 gateway (樹莓派) 和 server 建立獨立的帳號密碼。

THANK YOU